

Test Plan – Tic-Tac-Toe Web Application

Objective

Validate the functionality, usability, and reliability of the Tic-Tac-Toe application, including user management, gameplay, profile management, game history, and session handling.

Scope

User Management

- **Create a new player**
- **Select an existing player**
- **Logout functionality**

Gameplay

- **Start a new game**
- **Play against the computer**
- **Difficulty level selection**
- **Hint functionality**
- **Game result calculation**
- **New Game and Reset actions**

Profile

- **View player profile**
- **Update player name**
- **View game statistics summary**

History

- **View previously played games**
- **Verify recorded game details**
- **Clear game history**

Test Types

- **Functional Testing**
- **Exploratory Testing**
- **UI Validation**
- **End-to-End Testing**

Critical User Flows

Flow 1: New User Journey

Create Player → Start Game → Finish Game → Verify Profile → Verify History → Logout

Flow 2: Existing User Journey

Select Existing User → Play Game → Verify Statistics Update → Logout

Flow 3: Profile Management

Open Profile → Update Name → Verify Changes Persist

Flow 4: History Management

Play Multiple Games → Verify History Records → Clear History → Verify Removal

Flow 5: Gameplay Validation

Play Win/Lose/Draw Scenarios → Verify Results

Risks

- **AI logic may not respect selected difficulty.**
- **Hint functionality may provide incorrect recommendations.**
- **History may not accurately reflect played games.**
- **User profile updates may not persist correctly.**
- **Game state may become inconsistent after Reset/New Game actions.**

Entry Criteria

- **Application is accessible.**
- **Browser environment is available.**

Exit Criteria

- **All critical scenarios executed.**
- **Critical defects documented.**
- **Automation completed for critical flows.**